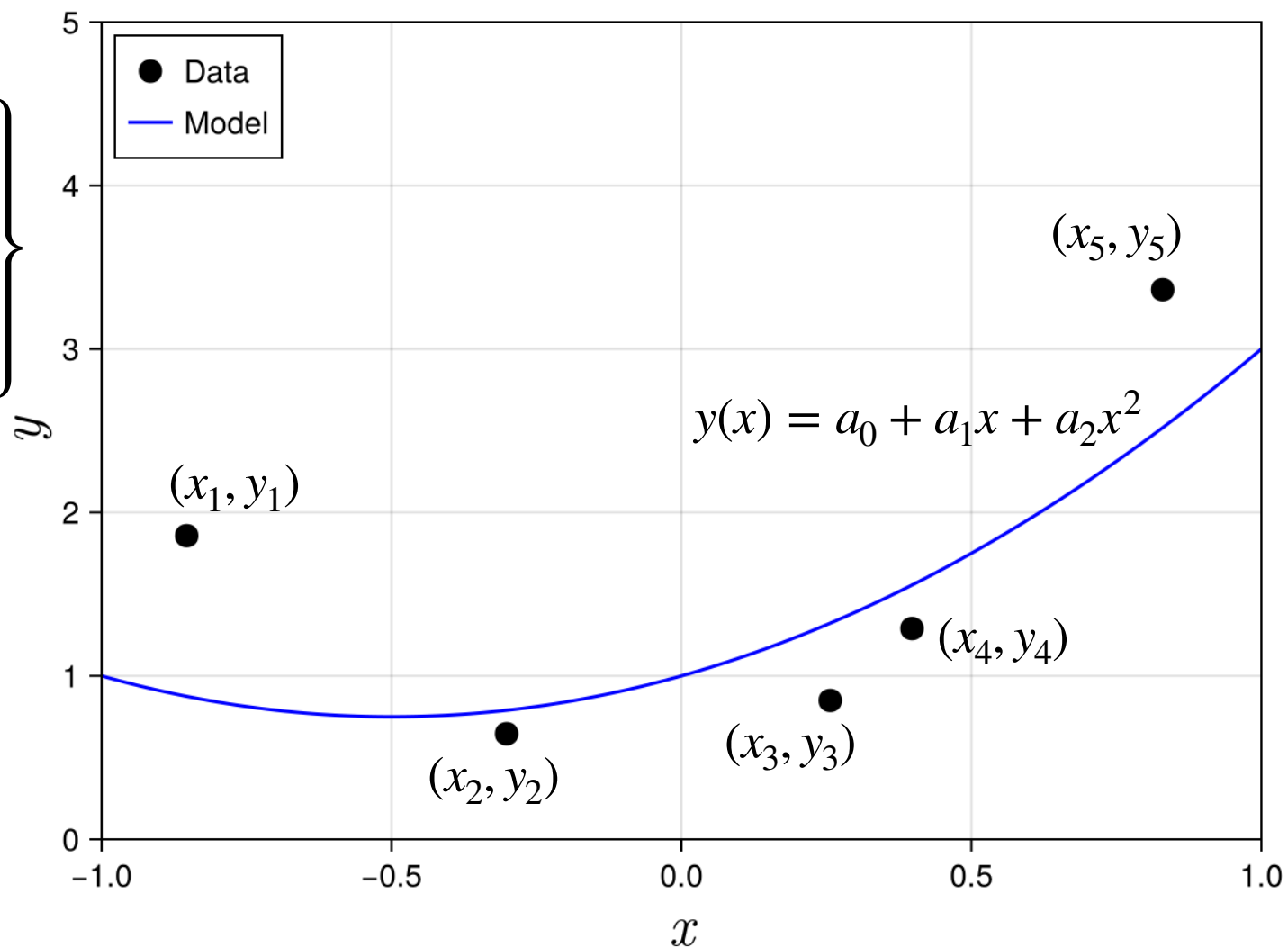


We want to find a_0, a_1 and a_2 such that

$$\begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \\ 1 & x_4 & x_4^2 \\ 1 & x_5 & x_5^2 \end{bmatrix} \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \end{Bmatrix} - \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{Bmatrix} \approx \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

or

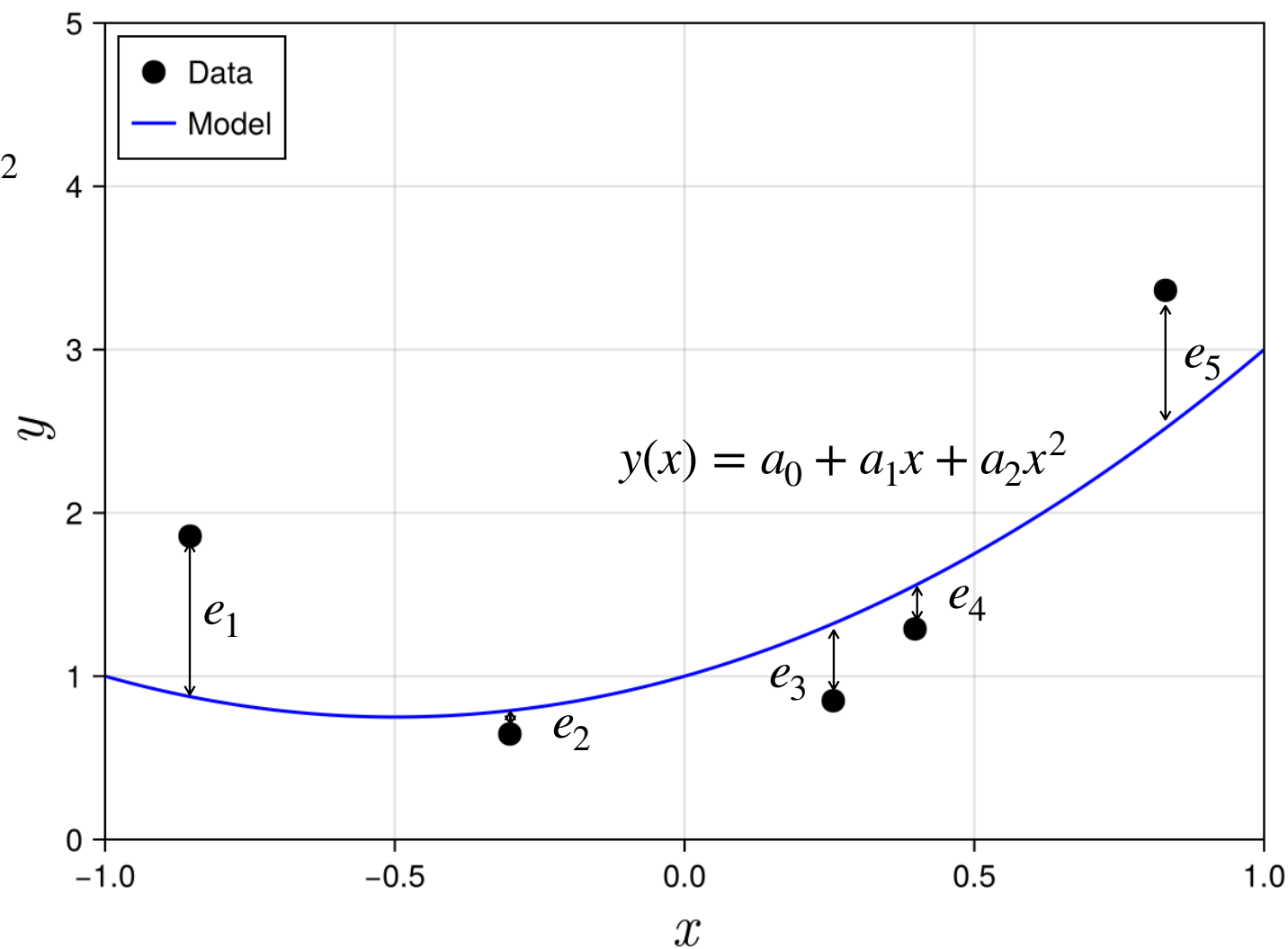
$$\begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \\ 1 & x_4 & x_4^2 \\ 1 & x_5 & x_5^2 \end{bmatrix} \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \end{Bmatrix} \approx \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{Bmatrix}$$



$$S = \sum_{i=1}^5 e_i^2$$

$$= \sum_{i=1}^5 (y_i - a_0 - a_1 x_i - a_2 x_i^2)^2$$

Want to minimize S



$$\frac{\partial S}{\partial a_0} = 0$$

$$a_0 \sum_{i=1}^5 + a_1 \sum_{i=1}^5 x_i + a_2 \sum_{i=1}^5 x_i^2 = \sum_{i=1}^5 y_i$$

$$\frac{\partial S}{\partial a_1} = 0$$

$$a_0 \sum_{i=1}^5 x_i + a_1 \sum_{i=1}^5 x_i^2 + a_2 \sum_{i=1}^5 x_i^3 = \sum_{i=1}^5 x_i y_i$$

$$\frac{\partial S}{\partial a_2} = 0$$

$$a_0 \sum_{i=1}^5 x_i^2 + a_1 \sum_{i=1}^5 x_i^3 + a_2 \sum_{i=1}^5 x_i^4 = \sum_{i=1}^5 x_i^2 y_i$$

$$\begin{bmatrix} 5 & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \end{Bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \sum x_i^2 y_i \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ x_1 & x_2 & x_3 & x_4 & x_5 \\ x_1^2 & x_2^2 & x_3^2 & x_4^2 & x_5^2 \end{bmatrix} \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \\ 1 & x_4 & x_4^2 \\ 1 & x_5 & x_5^2 \end{bmatrix} \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \end{Bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ x_1 & x_2 & x_3 & x_4 & x_5 \\ x_1^2 & x_2^2 & x_3^2 & x_4^2 & x_5^2 \end{bmatrix} \begin{Bmatrix} y_0 \\ y_1 \\ y_2 \end{Bmatrix}$$

$$V = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \\ 1 & x_4 & x_4^2 \\ 1 & x_5 & x_5^2 \end{bmatrix} \qquad [V]^T[V]\vec{a} = [V]^T\vec{y}$$

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V=hcat(ones(N_SAMPLES),x_samples,x_samples.^2)
a=V'V\'(V'y_samples)
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$$[V]^T[V]\vec{a} = [V]^T\vec{y}$$

$$\vec{a} = ([V]^T[V])^{-1}[V]^T\vec{y}$$

$$V = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \\ 1 & x_4 & x_4^2 \\ 1 & x_5 & x_5^2 \end{bmatrix}$$